

Number theory – 3

Quadratic residues and reciprocity

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Theoretical minimum. One question — is a a square modulo p — with a complete and computable answer. It is the standard source of obstructions for Diophantine equations.

Definitions. For an odd prime p and $p \nmid a$ the Legendre symbol is $\left(\frac{a}{p}\right) = 1$ if a is a quadratic residue modulo p and -1 otherwise. Exactly $(p-1)/2$ of the non-zero residues are squares — the image of the squaring map, which is two-to-one.

Theorem 1 (Euler’s criterion). $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$. *Proof:* the group $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic of order $p-1$ by Number theory – 2, and $a = g^k$ is a square exactly when k is even, which happens exactly when $a^{(p-1)/2} = g^{k(p-1)/2} = 1$. *Consequences:* the symbol is multiplicative, and $\left(\frac{-1}{p}\right) = 1$ if and only if $p \equiv 1 \pmod{4}$.

Theorem 2 (Gauss’s lemma and the second supplement). Let μ be the number of elements of $\{a, 2a, \dots, \frac{p-1}{2}a\}$ whose least positive residue exceeds $p/2$. Then $\left(\frac{a}{p}\right) = (-1)^\mu$. Counting μ for $a = 2$ gives $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$, that is 2 is a square modulo p exactly when $p \equiv \pm 1 \pmod{8}$.

Theorem 3 (quadratic reciprocity). For distinct odd primes p, q

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Proof (Eisenstein): by Gauss’s lemma each symbol is (-1) to the number of lattice points in a triangle, and the two triangles tile a rectangle with $\frac{p-1}{2} \cdot \frac{q-1}{2}$ lattice points. Together with the two supplements this computes any Legendre symbol as fast as the Euclidean algorithm; the Jacobi symbol $\left(\frac{a}{n}\right) = \prod \left(\frac{a}{p_i}\right)$ extends the computation to composite n , at the price that the value 1 no longer guarantees a square.

Theorem 4 (sums of two squares). A prime p is a sum of two squares if and only if $p = 2$ or $p \equiv 1 \pmod{4}$; a positive integer is a sum of two squares if and only if every prime $\equiv 3 \pmod{4}$ occurs in it to an even power. *Proof:* for $p \equiv 1 \pmod{4}$ take x with $x^2 \equiv -1$ and apply Thue's pigeonhole lemma, or factor in the Gaussian integers $\mathbb{Z}[i]$, which is Euclidean. The quadratic Gauss sum $\sum_a \left(\frac{a}{p}\right) e^{2\pi i a/p}$ equals $\sqrt{p}$ or $i\sqrt{p}$; it is evaluated in Fourier analysis – 4 and used in Trig-2.

1. Prove Euler's criterion and the multiplicativity of the Legendre symbol; determine $\left(\frac{-1}{p}\right)$ and deduce that there are infinitely many primes $\equiv 1 \pmod{4}$. Then prove Gauss's lemma and the formula for $\left(\frac{2}{p}\right)$.
2. Prove quadratic reciprocity by Eisenstein's lattice-point count, and use it to decide the solvability of $x^2 \equiv 219 \pmod{383}$ and to describe all primes p for which 3 is a quadratic residue.
3. Prove that for every odd prime p there is a natural number n such that the congruence $n^5 + n^4 - 3 \equiv x^2 \pmod{p}$ has no solution. (Mexmat 2013)
4. Let p be an odd prime and let $q(x) = \sum_{k=1}^{p-1} a_k x^k \in \mathbb{F}_p[x]$, where $a_k = k^{(p-1)/2} \pmod{p}$. Determine the largest power of $x - 1$ dividing $q(x)$ in $\mathbb{F}_p[x]$. (Putnam 2019)
5. Prove the two-squares theorem in both directions, by Thue's lemma and by factorisation in $\mathbb{Z}[i]$, and determine the number of representations of n as an ordered sum of two squares in terms of the divisors of n .